# Solow Model Cheatsheet

Macroeconomics I - Exam Reference

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# 1 Notation and Definitions

## Core Variables

| | | |
|---|---|---|
| $K$ = capital stock | $L$ = labor force | $Y$ = output |
| $A$ = technology (TFP) | $s$ = savings rate | |
| $\delta$ = depreciation rate | $n$ = population growth rate | $g$ = technology growth rate |

## Per-Worker Variables

$$k \equiv \frac{K}{L}, \qquad y \equiv \frac{Y}{L} = f(k)$$

## Per-Effective-Worker (Efficiency Unit) Variables

$$\tilde{k} \equiv \frac{K}{AL}, \qquad \tilde{y} \equiv \frac{Y}{AL} = f(\tilde{k})$$

## Constant Returns to Scale (CRS)

$$F(\lambda K, \lambda L) = \lambda F(K, L) \quad \forall \lambda > 0$$

## Fundamental Dynamic Equation

$$\dot{\tilde{k}} = sf(\tilde{k}) - (n + g + \delta)\tilde{k} \qquad \text{(Dynamic Equation)}$$

## Inada Conditions

$$\lim_{k \to 0} f'(k) = \infty, \qquad \lim_{k \to \infty} f'(k) = 0 \qquad \text{(Inada)}$$

## 2 Model Setup: Production Function

### Aggregate Production

$$Y = F(K, L) \quad \text{with Constant Returns to Scale} \tag{Production}$$

### Capital Accumulation

$$\dot{K} = sF(K, L) - \delta K \tag{Capital Accumulation}$$

### Per-Worker Form

$$\dot{k} = sf(k) - (n + \delta)k \tag{Per-Worker Dynamics}$$

### Per-Effective-Worker Form

$$\dot{\tilde{k}} = sf(\tilde{k}) - (n + g + \delta)\tilde{k} \tag{Per-Effective Dynamics}$$

**Break-Even Investment**

The term $(n + g + \delta)\tilde{k}$ represents *break-even investment*:

- $n\tilde{k}$: capital needed to equip new workers
- $g\tilde{k}$: capital needed for new effective units (technology growth)
- $\delta\tilde{k}$: replacement for depreciated capital

# 3 Derivation of Core Results

## 3.1 Derivation 1: Per-Worker Form

Starting with the capital accumulation equation:

$$\dot{K} = sF(K, L) - \delta K$$

Divide both sides by $L$ to get capital per worker:

$$\frac{\dot{K}}{L} = s\frac{F(K, L)}{L} - \delta\frac{K}{L}$$

Define $k = K/L$. Using the quotient rule for $\dot{k}$:

$$\dot{k} = \frac{\dot{K}L - K\dot{L}}{L^2} = \frac{\dot{K}}{L} - \frac{K}{L}\frac{\dot{L}}{L}$$

Since $\dot{L}/L = n$ (labor grows at rate $n$):

$$\dot{k} = \frac{\dot{K}}{L} - nk$$

Substituting $\dot{K}/L = sf(k) - \delta k$:

$$\dot{k} = sf(k) - \delta k - nk = sf(k) - (n + \delta)k$$

$$\boxed{\dot{k} = sf(k) - (n + \delta)k}$$

## 3.2 Derivation 2: Per-Effective-Worker Form

Starting with $\dot{K} = sF(K, L) - \delta K$
Define effective worker $\tilde{k} = K/(AL)$ where $A$ grows at rate $g$. Using product rule:

$$\dot{\tilde{k}} = \frac{\dot{K}}{AL} - \frac{K}{(AL)^2}(A\dot{L} + \dot{A}L)$$

Simplify the second term:

$$\dot{\tilde{k}} = \frac{\dot{K}}{AL} - \frac{K}{AL}\left(\frac{\dot{L}}{L} + \frac{\dot{A}}{A}\right) = \frac{\dot{K}}{AL} - \tilde{k}(n + g)$$

Now $\frac{\dot{K}}{AL} = s\frac{F(K,L)}{AL} - \delta\frac{K}{AL} = sf(\tilde{k}) - \delta\tilde{k}$:

$$\dot{\tilde{k}} = sf(\tilde{k}) - \delta\tilde{k} - \tilde{k}(n + g) = sf(\tilde{k}) - (n + g + \delta)\tilde{k}$$

Here: $sf(\tilde{k})$ is investment, $(n + g)\tilde{k}$ is dilution from labor and technology growth, $\delta\tilde{k}$ is depreciation.

$$\boxed{\dot{\tilde{k}} = sf(\tilde{k}) - (n + g + \delta)\tilde{k}}$$

### 3.3 Derivation 3: Steady-State

In the steady-state, effective capital per worker is constant:

$$\dot{\tilde{k}} = 0$$

Set the per-effective-worker accumulation equation to zero:

$$sf(\tilde{k}^*) - (n + g + \delta)\tilde{k}^* = 0$$

Rearrange to solve for steady-state capital:

$$sf(\tilde{k}^*) = (n + g + \delta)\tilde{k}^*$$

This equilibrium condition states that investment $sf(\tilde{k}^*)$ exactly equals break-even investment $(n + g + \delta)\tilde{k}^*$ needed to maintain $\tilde{k}^*$. At this point, all capital growth goes to replacing depreciated capital and equipping new effective workers.

$$\boxed{sf(\tilde{k}^*) = (n + g + \delta)\tilde{k}^*} \tag{SS}$$

**Steady-State Values:**

$$\tilde{y}^* = f(\tilde{k}^*) \tag{1}$$

$$\tilde{c}^* = \tilde{y}^* - (n + g + \delta)\tilde{k}^* \tag{2}$$

### 3.4 Derivation 4: Golden Rule

The steady-state consumption per effective worker is:

$$\tilde{c}^* = f(\tilde{k}^*) - (n + g + \delta)\tilde{k}^*$$

Maximize $\tilde{c}^*$ with respect to $\tilde{k}^*$ by taking the first-order condition:

$$\frac{d\tilde{c}^*}{d\tilde{k}^*} = f'(\tilde{k}^*) - (n + g + \delta) = 0$$

Rearrange:

$$f'(\tilde{k}^*_{GR}) = n + g + \delta$$

The Golden Rule capital $\tilde{k}^*_{GR}$ equates the marginal product of capital to the required rate of return $(n + g + \delta)$. At this point, the marginal unit of capital produces exactly enough to cover depreciation and population/technology growth, maximizing steady-state consumption.

$$\boxed{f'(\tilde{k}_{GR}) = n + g + \delta} \tag{GR}$$

# 4 Special Case: Cobb-Douglas

## Production Function

The Cobb-Douglas production function takes the form:

$$F(K, L) = K^{\alpha}(AL)^{1-\alpha}$$

where $\alpha \in (0, 1)$ is the capital share of income. The parameter $\alpha$ measures the elasticity of output with respect to capital.

**Interpretation of $\alpha$**

- $\alpha$ = capital share of income (elasticity of output w.r.t. capital)
- $1 - \alpha$ = labor share of income (elasticity of output w.r.t. labor)
- $\alpha$ typically ranges from 0.3 to 0.4 in empirical studies

## Per-Effective Worker Output

To express output in terms of capital per effective worker $\tilde{k} = K/(AL)$, we divide the production function by $AL$:

Derivation of $f(\tilde{k}) = \tilde{k}^{\alpha}$

$$\begin{aligned} f(\tilde{k}) &= \frac{F(K, L)}{AL} && (3) \\ &= \frac{K^{\alpha}(AL)^{1-\alpha}}{AL} && (4) \\ &= \left(\frac{K}{AL}\right)^{\alpha} && (5) \\ &= \tilde{k}^{\alpha} && (6) \end{aligned}$$

$$\boxed{f(\tilde{k}) = \tilde{k}^{\alpha}} \qquad \text{(CD-f)}$$

## Marginal Product of Capital

Derivation of MPK

$$\begin{aligned} f'(\tilde{k}) &= \frac{d}{d\tilde{k}}\tilde{k}^{\alpha} && (7) \\ &= \alpha\tilde{k}^{\alpha-1} && (8) \end{aligned}$$

$$\boxed{f'(\tilde{k}) = \alpha \tilde{k}^{\alpha-1}} \tag{CD-MPK}$$

Since $\alpha \in (0,1)$, we have $\alpha - 1 < 0$, which implies **diminishing returns to capital**.

## Steady-State Derivation

Derivation of Steady-State Capital

$$sf(\tilde{k}^*) = (n+g+\delta)\tilde{k}^* \tag{9}$$
$$s(\tilde{k}^*)^\alpha = (n+g+\delta)\tilde{k}^* \tag{10}$$
$$s\tilde{k}^{*\alpha-1} = (n+g+\delta) \tag{11}$$
$$\tilde{k}^* = \left[\frac{s}{n+g+\delta}\right]^{\frac{1}{1-\alpha}} \tag{12}$$

$$\boxed{\tilde{k}^* = \left[\frac{s}{n+g+\delta}\right]^{\frac{1}{1-\alpha}}} \tag{CD-SS-k}$$

### Steady-State Output and Consumption

$$\tilde{y}^* = \left[\frac{s}{n+g+\delta}\right]^{\frac{\alpha}{1-\alpha}} \tag{CD-SS-y}$$

$$\tilde{c}^* = \tilde{y}^* - (n+g+\delta)\tilde{k}^* \tag{CD-SS-c}$$

## Golden Rule Steady State

Derivation of Golden Rule Capital

$$\frac{d\tilde{c}^*}{d\tilde{k}} = 0 \tag{13}$$
$$\alpha\tilde{k}^{\alpha-1} - (n+g+\delta) = 0 \tag{14}$$
$$\tilde{k}_{GR}^{\alpha-1} = \frac{\alpha}{n+g+\delta} \tag{15}$$
$$\tilde{k}_{GR} = \left[\frac{\alpha}{n+g+\delta}\right]^{\frac{1}{1-\alpha}} \tag{16}$$

$$\boxed{\tilde{k}_{GR} = \left[\frac{\alpha}{n+g+\delta}\right]^{\frac{1}{1-\alpha}}} \quad \boxed{s_{GR} = \alpha} \qquad \text{(CD-GR)}$$

**Dynamic Efficiency**

| **If** $s < s_{GR}$ | **If** $s > s_{GR}$ |
|---|---|
| $\tilde{k}^* < \tilde{k}_{GR}$ | $\tilde{k}^* > \tilde{k}_{GR}$ |
| $\tilde{c}^* < \tilde{c}_{GR}$ | $\tilde{c}^* < \tilde{c}_{GR}$ |
| $f'(\tilde{k}^*) > n+g+\delta$ | $f'(\tilde{k}^*) < n+g+\delta$ |
| Increase $s$ to raise $\tilde{c}^*$ | Decrease $s$ to raise $\tilde{c}^*$ |
| Dynamically efficient | Dynamically inefficient |

## Convergence to Steady State

Derivation of Convergence Parameter Linearizing around the steady state:

$$\dot{\tilde{k}} \approx (1-\alpha)(n+g+\delta)(\tilde{k} - \tilde{k}^*)$$

At the steady state: $sf'(\tilde{k}^*) = (n+g+\delta)$, so:

$$\dot{\tilde{k}} = -(1-\alpha)(n+g+\delta)(\tilde{k} - \tilde{k}^*)$$

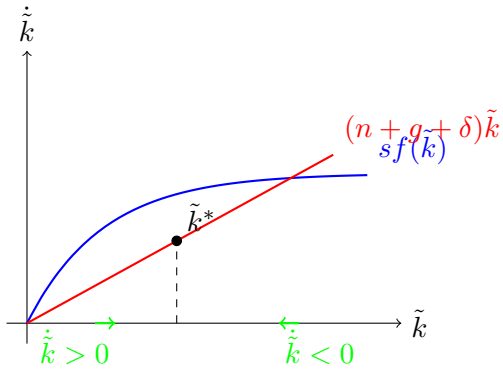
$$\boxed{\beta = (1-\alpha)(n+g+\delta)} \qquad \text{(CD-Convergence)}$$

**Properties of $\beta$**

- $\beta > 0$ since $0 < \alpha < 1$ and $n+g+\delta > 0$
- Higher $\alpha \Rightarrow$ slower convergence
- Higher $n$, $g$, or $\delta \Rightarrow$ faster convergence
- $s$ does NOT affect convergence speed in Cobb-Douglas
- Half-life: $t_{1/2} = \frac{\ln 2}{\beta}$

# 5 Dynamic Path, Stability, and Growth

## Phase Diagram




## Stability Analysis

- **For** $\tilde{k} < \tilde{k}^*$: $sf(\tilde{k}) > (n+g+\delta)\tilde{k} \Rightarrow \dot{\tilde{k}} > 0$ (capital accumulates)
- **For** $\tilde{k} > \tilde{k}^*$: $sf(\tilde{k}) < (n+g+\delta)\tilde{k} \Rightarrow \dot{\tilde{k}} < 0$ (capital depletes)

**Result:** The steady state $\tilde{k}^*$ is **globally asymptotically stable**.

## Transitional Dynamics

Log-linearization yields:

$$\gamma_{\tilde{k}} \equiv \frac{\dot{\tilde{k}}}{\tilde{k}} \approx -\beta(\ln \tilde{k} - \ln \tilde{k}^*)$$

Solution:

$$\ln(\tilde{k}_t) = (\ln(\tilde{k}_0) - \ln(\tilde{k}^*))e^{-\beta t} + \ln(\tilde{k}^*) \qquad \text{(Transition)}$$

## Conditional Convergence

The Solow model predicts **conditional convergence**: an economy grows faster the further it is from its own steady state.

# 6 Long-Run Growth and Growth Accounting

## Long-Run Growth

In the steady state:

- Per-effective variables $(\tilde{k}, \tilde{y}, \tilde{c})$ are constant
- Aggregate $Y$ grows at rate $n + g$
- Output per worker $y$ grows at rate $g + n$

## Growth Accounting

For Cobb-Douglas $Y = K^{\alpha}(AL)^{1-\alpha}$:

$$g_Y = \alpha g_K + (1 - \alpha)(g + n) \tag{Growth Accounting}$$

In steady state: $g_Y = g + n$.

## Comparative Statics on Steady State

| Parameter | Effect on $\tilde{k}^*$ | Effect on $\tilde{y}^*$ |
|---|---|---|
| $\uparrow s$ (savings) | $\uparrow$ | $\uparrow$ |
| $\uparrow n$ (population growth) | $\downarrow$ | $\downarrow$ |
| $\uparrow g$ (tech growth) | $\downarrow$ | $\downarrow$ |
| $\uparrow \delta$ (depreciation) | $\downarrow$ | $\downarrow$ |

# 7 Policy Shocks and Comparative Statics

## 4-Step Policy Shock Template

1. **Identify the parameter change:** What parameter changes? ($s$, $n$, $\delta$, $g$)
2. **Determine new steady state:** Solve with new parameter(s):

$$s' f(\tilde{k}^{*\prime}) = (n + g + \delta')\tilde{k}^{*\prime}$$

3. **Compute effects on outputs:**
   - $\tilde{y}^* = f(\tilde{k}^*)$
   - $\tilde{c}^* = \tilde{y}^* - (n + g + \delta)\tilde{k}^*$
4. **Describe transition path:**
   - If $\tilde{k}_0 < \tilde{k}^*_{\text{new}}$: economy accumulates ($\dot{\tilde{k}} > 0$)
   - If $\tilde{k}_0 > \tilde{k}^*_{\text{new}}$: economy decumulates ($\dot{\tilde{k}} < 0$)

# 8 Problem-Solving Templates

## (a) Derive Steady State and Trajectories

1. Start from fundamental law of motion: $\dot{\tilde{k}} = sf(\tilde{k}) - (n + g + \delta)\tilde{k}$
2. Set $\dot{\tilde{k}} = 0$ and solve: $sf(\tilde{k}^*) = (n + g + \delta)\tilde{k}^*$
3. For trajectories: log-linearize around steady state
4. Use phase diagram for qualitative analysis

## (b) Policy-Change Analysis

1. Identify parameter change ($s \to s'$, $\delta \to \delta'$, etc.)
2. Find new steady state with new parameters
3. Compare steady-state values $(\tilde{y}^*, \tilde{c}^*)$
4. Determine transition direction from initial $\tilde{k}_0$

## (c) Golden Rule with Given $f(\cdot)$

1. Set MPK equal to break-even investment: $f'(\tilde{k}_{GR}) = n + g + \delta$
2. Solve for $\tilde{k}_{GR}$ using the specific $f(\cdot)$ form
3. Compute $\tilde{c}_{GR} = f(\tilde{k}_{GR}) - (n + g + \delta)\tilde{k}_{GR}$

## (d) Per-Effective to Per-Worker Translation

1. Given $A(t) = A_0 e^{gt}$, $L(t) = L_0 e^{nt}$
2. Per-effective: $\tilde{k} = K/(AL)$, $\tilde{y} = Y/(AL) = f(\tilde{k})$
3. Per-worker: $k = K/L = \tilde{k} \cdot A$, $y = Y/L = \tilde{y} \cdot A$
4. In SS: $\tilde{k}$ constant $\Rightarrow$ $k$ grows at rate $g$, $y$ at $g + n$

## (e) Non-Cobb-Douglas Production

1. Identify $f(\tilde{k})$ from $F(K, AL)$: $f(\tilde{k}) = F(\tilde{k}, 1)$
2. Apply general SS condition: $sf(\tilde{k}^*) = (n + g + \delta)\tilde{k}^*$
3. For comparative statics: differentiate implicitly
4. Verify uniqueness with Inada conditions

# 9 Common Pitfalls and Tips

1. **Per-effective vs per-worker mix-ups**
   - $\tilde{k} = K/(AL)$ vs $k = K/L$
   - Dynamic equation: $\dot{\tilde{k}}$ uses $(n + g + \delta)$, $\dot{k}$ uses $(n + \delta)$
   - In SS: $\tilde{k}$ constant, $k$ grows at rate $g$
2. **Mistaking steady-state conditions**
   - $\dot{k} = 0$ means $k$ is constant (not growing) $\dot{\tilde{k}} = 0$ means $\tilde{k}$ is constant
3. **Misapplying Golden Rule**
   - Golden Rule maximizes $\tilde{c}^*$, not $\tilde{y}^*$
   - Condition: $f'(\tilde{k}^*) = n + g + \delta$
   - Under CD: $s_{GR} = \alpha$, not $s = 1$
4. **Neglecting $A$-growth**
   - $g$ affects steady state $\tilde{k}^*$: higher $g \Rightarrow$ lower $\tilde{k}^*$
5. **Convergence speed misconceptions**
   - $\beta = (1 - \alpha)(n + g + \delta)$, NOT $(1 - \alpha)$

## 10 Quick Reference Formulas

### Core Variables

$$\tilde{k} = \frac{K}{AL}, \quad \tilde{y} = \frac{Y}{AL} = f(\tilde{k})$$
$$k = \frac{K}{L}, \quad y = \frac{Y}{L} = f(k)$$

### Dynamic Equations

$$\dot{\tilde{k}} = sf(\tilde{k}) - (n + g + \delta)\tilde{k}$$
$$\dot{k} = sf(k) - (n + \delta)k$$

### Steady State

$$sf(\tilde{k}^*) = (n + g + \delta)\tilde{k}^*$$
$$\tilde{y}^* = f(\tilde{k}^*)$$
$$\tilde{c}^* = \tilde{y}^* - (n + g + \delta)\tilde{k}^*$$

### Cobb-Douglas ($f(\tilde{k}) = \tilde{k}^\alpha$, $0 < \alpha < 1$)

$$\tilde{k}^* = \left[\frac{s}{n + g + \delta}\right]^{\frac{1}{1-\alpha}}$$
$$\tilde{y}^* = \left[\frac{s}{n + g + \delta}\right]^{\frac{\alpha}{1-\alpha}}$$
$$\tilde{k}_{GR} = \left[\frac{\alpha}{n + g + \delta}\right]^{\frac{1}{1-\alpha}}$$
$$s_{GR} = \alpha$$

### Growth Rates in SS

- Per-effective $(\tilde{k}, \tilde{y}, \tilde{c})$: constant (0 growth)
- $k$ grows at rate $g$
- $y$ grows at rate $g + n$
- $Y$ grows at rate $g + n$

### Convergence

$$\beta = (1 - \alpha)(n + g + \delta), \quad \ln(\tilde{k}_t) = (\ln(\tilde{k}_0) - \ln(\tilde{k}^*))e^{-\beta t} + \ln(\tilde{k}^*)$$

### Growth Accounting

$$g_Y = \alpha g_K + (1 - \alpha)(g + n)$$

# 11 Implementation Notes

## Changes from Original Version

1. **Enhanced Preamble**: Added tcolorbox, siunitx, cleveref, and hyperref packages with proper configuration
2. **Derivation Boxes**: Created dedicated derivation environments (derivationbox, resultbox) using tcolorbox for clear visual separation
3. **Narrative Derivations**: Added step-by-step derivations with inline commentary for:
   - Per-worker to per-effective transformation
   - Steady-state derivation
   - Golden Rule derivation
   - Cobb-Douglas closed forms
   - Convergence speed
4. **Equation Numbering**: Used `tag` for consistent equation references
5. **Visual Hierarchy**: Added colored boxes for key interpretations and properties
6. **Consistent Notation**: Used $\tilde{k}$, $\tilde{y}$, $\tilde{c}$ commands throughout

## Packages Used

- amsmath, amssymb, mathtools: Mathematics
- graphicx: Graphics
- xcolor: Color support
- siunitx: Units and number formatting
- cleveref: Smart cross-references
- hyperref: Hyperlinks
- tcolorbox: Colored boxes for derivations
- booktabs: Publication-quality tables

## Compilation

Compile with: `pdflatex` or `lualatex`